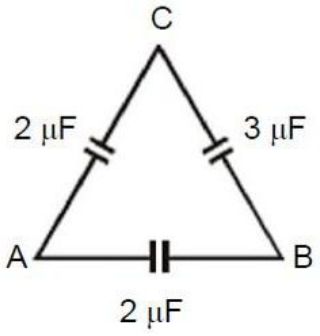
21 Three capacitors are connected in the shape of a triangle ABC as shown in the figure below.



A 5.0 V battery is connected between A and B.

What is the voltage between B and C?

A 1.0 V **B** 2.0 V **C** 2.5 V **D** 3.0 V

Ans: B

Since the 2 μF (AB) capacitor is in parallel with the series combination of 2 μF and 3 μF, the voltage across the entire AC-CB branch is also **5.0 V**.

For capacitors in series, the equivalent capacitance is:
$1/C_{eq} = 1/2 + 1/3 = 5/6 \rightarrow C_{eq} = 1.2\ \mu F$

Charge on each capacitor in the series branch:
$Q = C_{eq}V = 1.2 \times 5.0 = 6.0\ \mu C$

$V_{BC} = Q/C_{BC} = 6.0/3 =$ **2.0 V**